

Networks Decide for Us

Imputed homophilic networks and the structural premium of diffusion

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The question

- Collective behaviors (innovations, mobilizations) spread as **adoption cascades** on social networks.
- Most models use *synthetic* topologies and treat agents as simple relays [1].
- We cluster with people like us. Real networks are shaped by **multidimensional homophily** [2].
- Individual *preferences* to adopting behaviors **can be measured** in surveys.
- Studies often uses peer influence as the only mechanism for adoption.

Where we go

Two moves

- **Impute** plausible whole-networks onto real survey respondents from empirical homophily.
- **Measure** the causal premium of that structure on diffusion.

Conditional on *fixed individual characteristics*, does collective behavior depend on the network's **shape**, or on **who sits next to whom**?

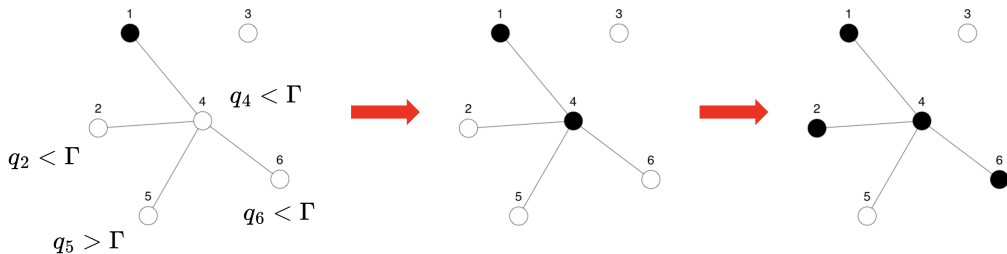
The model

Adoption, part 1 — Rational Choice

- A behavior has an *Intrinsic Utility Level* **IUL**, $\Gamma \in [0, 1]$.
- Each person has a *Minimum Utility Requirement* **MUR**, $q_i \in [0, 1]$, representing their aversion to adopting the behavior.

An agent adopts on its own iff

$$q_i \leq \Gamma \implies a_i = 1$$

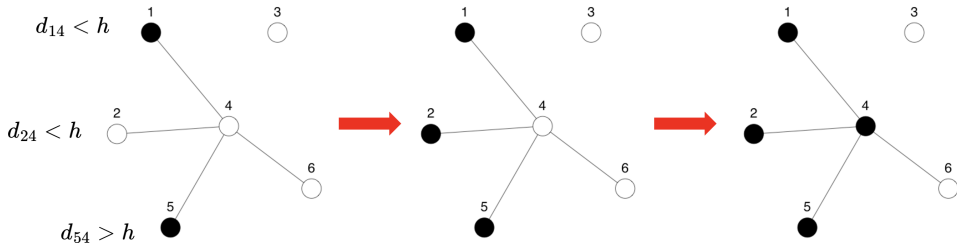


Adoption, part 2 — Selective Social Influence

Adopt by peer contagion if $\tau_i \leq \tilde{E}_i$ [5,6] — but only those socially *close* count:

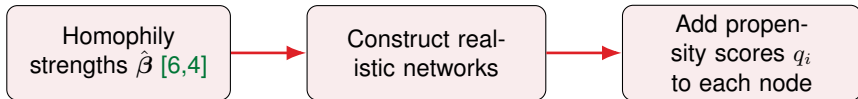
$$\tilde{E}_i \equiv \frac{\sum_{j \neq i} \mathbf{X}_{ij} \tilde{a}_j}{\sum_{j \neq i} \mathbf{X}_{ij}}, \quad \tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma(MSP - d_{ij})$$

- There is a Maximum Social Proximity **MSP** where the limit of influence sits [7].



Imputing structure & running the model

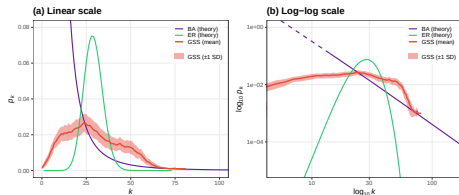
Imputing a network onto a survey



Tie probability is a function of social distance (age, sex, education, race, religion):

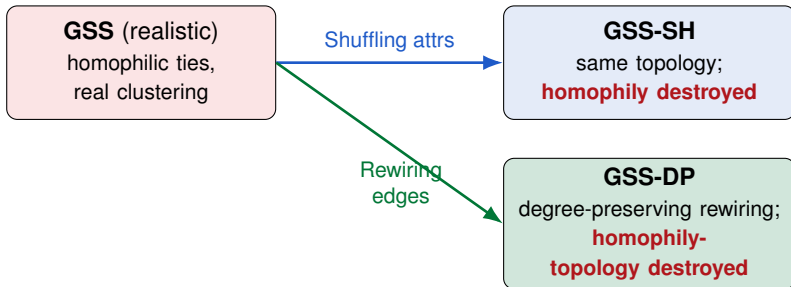
$$\text{logit } \Pr(Y_{ij} = 1) = \alpha + \beta^\top \mathbf{d}_{ij}$$

$\hat{\beta}$ comes from GSS confidant data [6]; propensities q_i come from real survey answers (innovation, collective action).



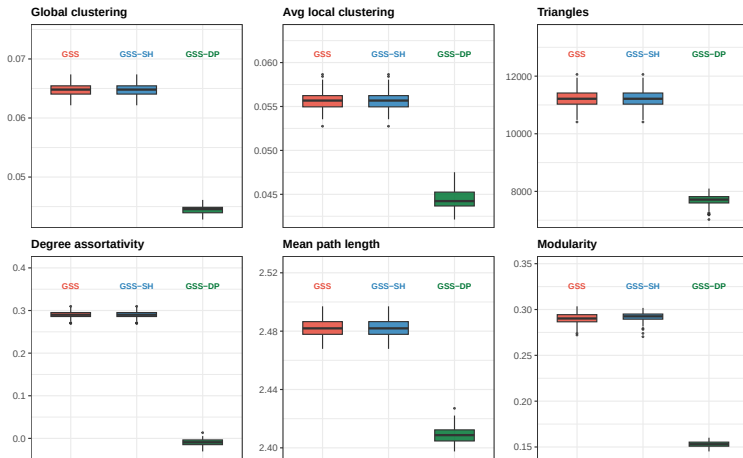
Right-skewed degree structure that Erdős-Rényi / Barabási-Albert models fail to match jointly.

Two ways to “break” the network



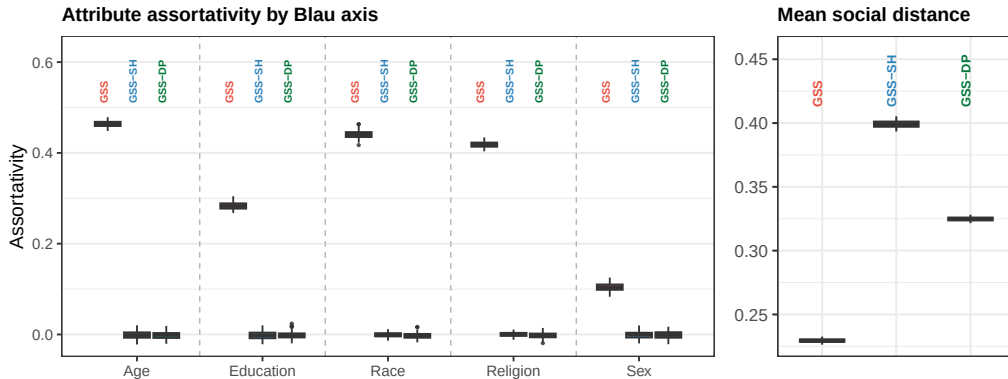
GSS vs *GSS-SH* isolates **homophily**; *GSS-SH* vs *GSS-DP* how much **topology** adds.

What is destroyed? — topology



Topology statistics across 100 networks each. **GSS** and **GSS-SH** are identical; only **GSS-DP** degrades them: clustering -31% , triangles -31% , modularity -47% .

What is destroyed? — homophily



both GSS-SH and GSS-DP annihilate homophily.

So the shuffle destroys *homophily only*, the rewiring destroys *both*.

Comprehensive simulation design

Hybrid social contagion via `netdiffuseR` [7], over plausible **GSS** vs. **GSS-DP** (degree-preserving) and **GSS-SH** (shuffled attributes) networks.

- **IUL** (Γ): 41 levels $[0, 1]$. **MSP** (h): 13 levels $[0, 1]$.
- **Thresholds** $\tau_i \sim \mathcal{N}(\mu, \sigma)$: $\mu \in \{.3, .4, .5, .6\}$, $\sigma \in \{.08, .12, .16, .20\}$.
- **Seeding**: random, central, marginal, closeness, eigenvector.

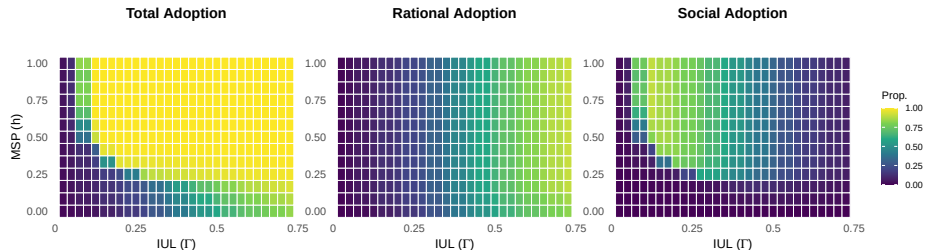
Total: $41 \times 13 \times 4 \times 4 \times 5 \times (\text{runs}) \approx 4 \times 10^6$ **simulations**

We then fit *Big Additive Models* (BAM) to isolate the structural effect:

$$\text{logit}[\mathbb{E}(\Phi)] = \beta_0 + \mathbf{Z}\boldsymbol{\beta} + \beta_{\text{DP}} \mathbb{I}_{\text{DP}} + \beta_{\text{SH}} \mathbb{I}_{\text{SH}} + s(\text{IUL}, \text{MSP})$$

Results

Result 1 — How adoption plays out



Mean adoption over the attractiveness (IUL) – openness (MSP) plane.

Rational adoption grows with attractiveness; **social** adoption is more complex, so total adoption stays high over a wide region *even when the behavior is barely attractive*.

Result 2 — Breaking structure cuts adoption

How are the adoption results when we compare with the broken versions?

vs. GSS	OR	odds drop
GSS-SH (shuffle)	0.465	-53%
GSS-DP (rewire)	0.432	-57%

Same individuals, same propensities, but adoption falls by more than half.

In both the *homophily is destroyed* - lack of **echo chambers**

Total adoption (GSS)	Estimate	OR
(Intercept)	6.384***	—
τ_{mean}	-6.784***	—
τ_{sd}	5.820***	—
<i>Seeding (ref: Random)</i>		
Central	0.130***	1.139
Closeness	0.130***	1.138
Eigenvector	0.129***	1.138
Marginal	-0.073***	0.930
<i>Network (ref: GSS)</i>		
GSS-SH	-0.766***	0.465
GSS-DP	-0.840***	0.432
Dev. expl.: 82.6% $N \approx 5.1 \times 10^6$		

*** $p < 0.001$. OR = $\exp(\beta)$.

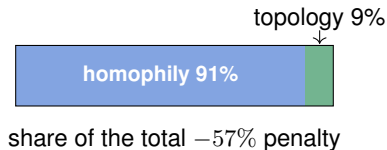
Result 2 — Homophily alone reproduces most of the drop

The two models compare like this:

GSS-SH -53% homophily destroyed
GSS-DP -57% homophily *and* topology

Breaking *only* homophily — keeping the topology — already recovers ~**91%** of the full drop.

⇒ **homophily is doing most of the work.**



Conclusions

1. **A method:** you can build **realistic**, simulable whole-networks for survey respondents out of **empirical homophily**.
2. **Three findings:**
 1. Social influence explains adoption even for non-attractive behaviors*.
 2. Empirical structure yields a **+57% adoption premium**.
 3. In realistic social networks, topology it not all: **who sits next to whom** is critical.
3. **A mechanism:** what's doing the work is **homophily**, not the abstract topology.

*Where a behavior ends up is not the sum of individual preferences,
but it's **governed by the structure of our real ties**.*

References

1. McPherson, Smith-Lovin & Cook (2001). Birds of a Feather. *Annu. Rev. Sociol.* 27.
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Thanks!



Scan: preprint draft (Zenodo)

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<https://doi.org/10.5281/zenodo.19360209>

Backup — the *shape* of change also differs

Beyond *how much*, structure governs *how* adoption tips.

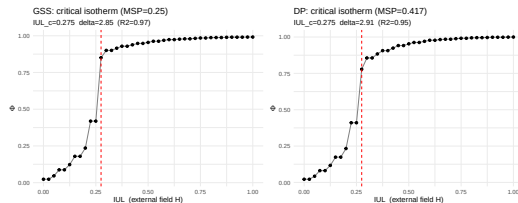
$$\chi \sim |MSP - MSP_c|^{-\gamma}$$

$$\Phi \sim |IUL - IUL_c|^{-\delta}$$

And here the *other* structural property does the work: **GSS-SH keeps universality** ($\gamma \approx 1$) even though it has no homophily — only **GSS-DP**, which destroys the *topology*, loses it [9].

Suscept. γ	GSS	GSS-SH	GSS-DP
mean	≈ 0.91	≈ 1.08	≈ 3.9

Double dissociation: homophily $\rightarrow$ *volume* (Result 2);
topology $\rightarrow$ *continuity* (here).



Critical isotherms $\Phi(IUL)$ at $MSP=MSP_c$: along the field axis *both* give $\delta \approx 3$; only γ (the openness axis) discriminates.

Technical details

- **Social distance** $d_{ij} \in [0, 1]$ — Gower distance over the Blau-space attributes (race, sex, religion, age, education):

$$d_{ij} = \frac{1}{p} \sum_{k=1}^p \delta_{ij}^k, \quad \delta_{ij}^k = \begin{cases} \mathbb{I}[x_i^k \neq x_j^k] & \text{categorical (race, sex, religion)} \\ \frac{|x_i^k - x_j^k|}{\max_k - \min_k} & \text{numeric (age, education)} \end{cases}$$

* The idiosyncrasy that drives this is **measured preference** and **realistic structure**, not stochastic noise (Macy & Evtushenko [8]).